

lab05_leila_kawanishi_277185

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Laboratório Especial

Atividade

Ao longo desta semana, você deverá desenvolver código em R, utilizando-se de tidyverse, ggplot, leaflet, ggcal e outros pacotes disponibilizados em HTML Widgets para responder uma questão relacionada ao conjunto de dados flights. Para tanto, considere os seguintes fatos:

- Cias Aéreas (AIRLINE) possuem diferentes aeronaves (codificadas em TAIL_NUMBER);
- (Exemplo hipotético) No dia 01/01/2015, a aeronave N431WN viaja de BWI para JFK;
- (Exemplo hipotético) No dia 02/01/2015, a aeronave N431WN viaja de JFK para ORD;
- E assim sucessivamente. . .

No caso apontado acima, pode-se ver que a aeronave N431WN realizou um trajeto (provavelmente único) durante todo o ano de 2015. Nos dias exemplificados, o trajeto foi BWI-JFK-ORD.

Crie uma função que:

- Receba um valor de TAIL_NUMBER (por exemplo, N431WN);
- Produza uma tabela (tidy) com todos os trajetos realizados pela aeronave (ordenadas por data e hora, contendo todas as colunas do arquivo flights.csv.zip);
- Produza um mapa que apresente todo o trajeto voado pela aeronave ao longo de todo o ano; o trajeto deve ser apresentado de maneira linear no tempo (i.e., segue a sequência do tempo, como no exemplo hipotético dado acima);
- O mapa deve ser decorado com estatísticas do seu interesse (por exemplo, velocidade média do voo como espessura da linha que conecta os aeroportos envolvidos no trajeto);

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2     4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr       1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ggplot2)
library(leaflet) #creates interactive maps
library(ggcal)#creats calendars
```

```
#to examine the table before preparing the section-by-section reading
flights_amostra <- read_csv("../flights.csv",
                             n_max = 5)
```

Rows: 5 Columns: 31

```
-- Column specification -----
Delimiter: ","
chr (10): AIRLINE, TAIL_NUMBER, ORIGIN_AIRPORT, DESTINATION_AIRPORT, SCHEDUL...
dbl (15): YEAR, MONTH, DAY, DAY_OF_WEEK, FLIGHT_NUMBER, DEPARTURE_DELAY, TAX...
lgl (6): CANCELLATION_REASON, AIR_SYSTEM_DELAY, SECURITY_DELAY, AIRLINE_DEL...
```

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
glimpse(flights_amostra)
```

Rows: 5

Columns: 31

```
$ YEAR      <dbl> 2015, 2015, 2015, 2015, 2015
$ MONTH     <dbl> 1, 1, 1, 1, 1
$ DAY       <dbl> 1, 1, 1, 1, 1
$ DAY_OF_WEEK <dbl> 4, 4, 4, 4, 4
$ AIRLINE    <chr> "AS", "AA", "US", "AA", "AS"
$ FLIGHT_NUMBER <dbl> 98, 2336, 840, 258, 135
$ TAIL_NUMBER <chr> "N407AS", "N3KUAA", "N171US", "N3HYAA", "N527AS"
$ ORIGIN_AIRPORT <chr> "ANC", "LAX", "SFO", "LAX", "SEA"
```

```

$ DESTINATION_AIRPORT <chr> "SEA", "PBI", "CLT", "MIA", "ANC"
$ SCHEDULED_DEPARTURE <chr> "0005", "0010", "0020", "0020", "0025"
$ DEPARTURE_TIME <chr> "2354", "0002", "0018", "0015", "0024"
$ DEPARTURE_DELAY <dbl> -11, -8, -2, -5, -1
$ TAXI_OUT <dbl> 21, 12, 16, 15, 11
$ WHEELS_OFF <chr> "0015", "0014", "0034", "0030", "0035"
$ SCHEDULED_TIME <dbl> 205, 280, 286, 285, 235
$ ELAPSED_TIME <dbl> 194, 279, 293, 281, 215
$ AIR_TIME <dbl> 169, 263, 266, 258, 199
$ DISTANCE <dbl> 1448, 2330, 2296, 2342, 1448
$ WHEELS_ON <chr> "0404", "0737", "0800", "0748", "0254"
$ TAXI_IN <dbl> 4, 4, 11, 8, 5
$ SCHEDULED_ARRIVAL <chr> "0430", "0750", "0806", "0805", "0320"
$ ARRIVAL_TIME <chr> "0408", "0741", "0811", "0756", "0259"
$ ARRIVAL_DELAY <dbl> -22, -9, 5, -9, -21
$ DIVERTED <dbl> 0, 0, 0, 0, 0
$ CANCELLED <dbl> 0, 0, 0, 0, 0
$ CANCELLATION_REASON <lg1> NA, NA, NA, NA, NA
$ AIR_SYSTEM_DELAY <lg1> NA, NA, NA, NA, NA
$ SECURITY_DELAY <lg1> NA, NA, NA, NA, NA
$ AIRLINE_DELAY <lg1> NA, NA, NA, NA, NA
$ LATE_AIRCRAFT_DELAY <lg1> NA, NA, NA, NA, NA
$ WEATHER_DELAY <lg1> NA, NA, NA, NA, NA

```

```
#Creanting the function
```

```
analisa_aeronave <- function(tail_number, arquivo){ #accepts arguments "tail_number" e "arquivo"
```

```
#select the aircraft
```

```
  pega_aeronave <- function(data, pos){
```

```
    data %>% # "data" refers to each piece that read_csv_chunked deliver
```

```
    filter(TAIL_NUMBER == tail_number) #retains only the aircraft passed to the function
```

```
  }
```

```
  callback <- DataFrameCallback$new(pega_aeronave) #for each piece you read, execute "pega_aeronave"
```

```
#We read the file in chunks
```

```
  voos <- read_csv_chunked(arquivo,
    callback = callback,
```

```

        chunk_size = 100000
    )

#create and organized date and time
voos <- voos %>%
mutate(
  data = as.Date(
    paste(YEAR, MONTH, DAY, sep = "-")
  ),

  hora = paste0(
    substr(DEPARTURE_TIME, 1, 2),
    ":",
    substr(DEPARTURE_TIME, 3, 4)
  ),

  data_hora = as.POSIXct(
    paste(data, hora),
    format = "%Y-%m-%d %H:%M"
  )
) %>%
arrange(data_hora)

#loads database airports

aeroportos <- read_csv("../airports.csv")

#select the coordinators
aeroportos_coord <- aeroportos %>%
  select(IATA_CODE, LATITUDE, LONGITUDE) #selects only the columns of interest

#renames the origin and destination latitudes and longitudes
aero_origem <- aeroportos_coord %>%
  rename(ORIGIN_LAT = LATITUDE,
         ORIGIN_LON = LONGITUDE
  )
aero_destino <- aeroportos_coord %>%
  rename(DEST_LAT = LATITUDE,
         DEST_LON = LONGITUDE
  )

```

```

#combines flights.csv table with the coordinates of the origin and destination from de airport
voos <- voos %>%
  left_join(aero_origem,
            by = c("ORIGIN_AIRPORT" = "IATA_CODE")) %>%
  left_join(aero_destino,
            by = c("DESTINATION_AIRPORT" = "IATA_CODE"))

#calculates speed
voos <- voos %>%
  mutate(
    VELOCIDADE = DISTANCE / (AIR_TIME / 60)
  )

#creates map
mapa <- leaflet() %>%
  addTiles()

#draw each flight
for (i in 1:nrow(voos)) {

  mapa <- mapa %>%
    addPolylines(
      lat = c(voos$ORIGIN_LAT[i], voos$DEST_LAT[i]),
      lng = c(voos$ORIGIN_LON[i], voos$DEST_LON[i]),
      weight = voos$VELOCIDADE[i] / 100
    )
}

#return table and map
return(
  list(
    tabela = voos,
    mapa = mapa
  )
)
}

```

```
resultado <- analisa_aeronave("N431WN", "../flights.csv")
```

```
-- Column specification -----  
cols(  
  .default = col_double(),  
  AIRLINE = col_character(),  
  TAIL_NUMBER = col_character(),  
  ORIGIN_AIRPORT = col_character(),  
  DESTINATION_AIRPORT = col_character(),  
  SCHEDULED_DEPARTURE = col_character(),  
  DEPARTURE_TIME = col_character(),  
  WHEELS_OFF = col_character(),  
  WHEELS_ON = col_character(),  
  SCHEDULED_ARRIVAL = col_character(),  
  ARRIVAL_TIME = col_character(),  
  CANCELLATION_REASON = col_character()  
)  
i Use `spec()` for the full column specifications.
```

```
Rows: 322 Columns: 7
```

```
-- Column specification -----  
Delimiter: ","  
chr (5): IATA_CODE, AIRPORT, CITY, STATE, COUNTRY  
dbl (2): LATITUDE, LONGITUDE
```

```
i Use `spec()` to retrieve the full column specification for this data.  
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
names(resultado)
```

```
[1] "tabela" "mapa"
```

```
dim(resultado$tabela)
```

```
[1] 1938  39
```

```
resultado$tabela
```

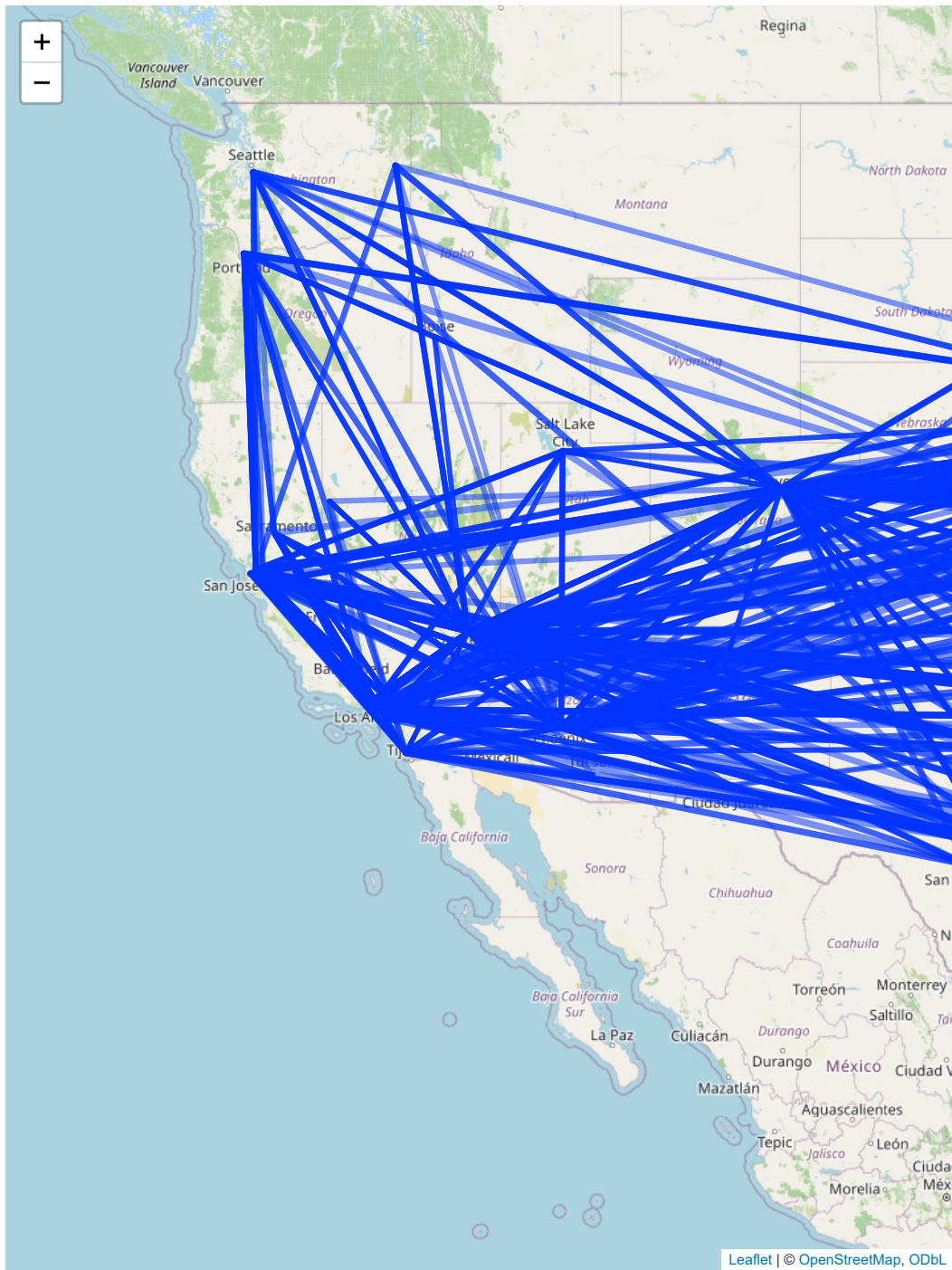
```
# A tibble: 1,938 x 39
```

```
  YEAR MONTH   DAY DAY_OF_WEEK AIRLINE FLIGHT_NUMBER TAIL_NUMBER
  <dbl> <dbl> <dbl>      <dbl> <chr>          <dbl> <chr>
1  2015     1     1          4 WN             1762 N431WN
2  2015     1     1          4 WN             3564 N431WN
3  2015     1     1          4 WN             3564 N431WN
4  2015     1     1          4 WN             3564 N431WN
5  2015     1     2          5 WN             1793 N431WN
6  2015     1     2          5 WN             1793 N431WN
7  2015     1     2          5 WN             1793 N431WN
8  2015     1     2          5 WN             1793 N431WN
9  2015     1     2          5 WN             1793 N431WN
10 2015     1     2          5 WN             3144 N431WN
```

```
# i 1,928 more rows
```

```
# i 32 more variables: ORIGIN_AIRPORT <chr>, DESTINATION_AIRPORT <chr>,
# SCHEDULED_DEPARTURE <chr>, DEPARTURE_TIME <chr>, DEPARTURE_DELAY <dbl>,
# TAXI_OUT <dbl>, WHEELS_OFF <chr>, SCHEDULED_TIME <dbl>, ELAPSED_TIME <dbl>,
# AIR_TIME <dbl>, DISTANCE <dbl>, WHEELS_ON <chr>, TAXI_IN <dbl>,
# SCHEDULED_ARRIVAL <chr>, ARRIVAL_TIME <chr>, ARRIVAL_DELAY <dbl>,
# DIVERTED <dbl>, CANCELLED <dbl>, CANCELLATION_REASON <chr>, ...
```

```
resultado$mapa
```



```
Sys.setenv(TZ = "America/Sao_Paulo")

paste(
  "Arquivo compilado em:",
  format(Sys.time(), "%d/%m/%Y às %H:%M:%S")
)
```

```
[1] "Arquivo compilado em: 12/09/2026 às 01:12:54"
```